Hammad Ahmad

AI / Machine Learning Engineer | LLMs · RAG · Knowledge Graphs

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LinkedIn | GitHub | Portfolio | ORCID

PROFILE

AI/ML engineer with an MSc (Merit) in Applied AI and a first-author peer-reviewed publication (Springer). I build RAG systems pairing vector embeddings with knowledge graphs, benchmark predictive models on large real-world datasets, and work across the stack, from retrieval design to backend APIs and deployment.

TECHNICAL SKILLS

Machine Learning & NLP : PyTorch, TensorFlow, scikit-learn, LLMs, Retrieval-Augmented Generation (RAG), Sentence Transformers, MLflow

LLM & Retrieval : Vector embeddings, semantic search, cross-encoder re-ranking, RAGAS evaluation, Neo4j knowledge graphs, Ollama

Engineering : Python, C++, SQL, FastAPI, Flask, REST APIs, React

Data & Infrastructure : PostgreSQL, Neo4j, pandas, NumPy, Docker, Git, Linux

EXPERIENCE

Outlyst — Oct 2025 - Mar 2026
AI / Machine Learning Engineer — Leeds, UK (Remote)

- Built backend dialogue-flow logic and workflow automation for an AI voice-agent system (Retell AI, FastAPI), integrating LLM-based dialogue with production infrastructure.
- Developed asynchronous backend services and lead-tracking automation for an AI-driven outbound communication platform.

University of Bradford — Jan 2025 - Sep 2025
Research Assistant, Machine Learning & LLMs — Bradford, UK

- Built FinLaw-UK, a RAG system pairing Mistral 7B (served locally via Ollama) with a Neo4j knowledge graph for UK financial-regulation question answering, grounded in FCA, PRA, FRC and statutory sources.
- Engineered the retrieval pipeline: clause-level segmentation, Sentence Transformer embeddings, cross-encoder re-ranking, and Neo4j citation checks that flag missing references as hallucinations.
- Evaluated on a 110-item legal QA set with RAGAS and custom metrics: 0.82 source accuracy and 0.81 citation quality, improving on a standalone LLM baseline.

COMSATS University Islamabad — Jul 2023 - Jul 2024
Research Assistant, Data Science — Islamabad, Pakistan

- Benchmarked 11 classifiers (Random Forest, XGBoost, LightGBM, CatBoost and others) for diabetes risk on a 253,680-record CDC BRFSS dataset; Random Forest reached 93.2% accuracy and 0.99 AUC after correcting severe class imbalance with random over-sampling.
- Deployed the model as a risk-screening web app (React, Flask REST API) returning personalised risk probabilities.

PROJECTS

Jobzyl: Multi-Platform Job Aggregator — jobzyl.com
Full-Stack Architecture, Data Pipelines, Responsive UI

- Built a full-stack platform that ingests real-time job listings from multiple sources (LinkedIn, Indeed, Google Jobs) into a single searchable portal.
- Designed the data-parsing pipelines and responsive UI; live and actively maintained.

EDUCATION

University of Bradford — Sep 2024 - Sep 2025
MSc, Applied Artificial Intelligence and Data Analytics (Merit) — Bradford, UK

- **Dissertation**: FinLaw-UK: A Graph-Augmented Retrieval Chatbot for Reliable and Transparent UK Financial Regulation.

COMSATS University Islamabad — Sep 2020 - Jul 2024
BS, Bioinformatics — Islamabad, Pakistan

PUBLICATIONS

Ahmad, H., Khan, M.U., & Azam, M. (2024). Comparative Analysis of Machine Learning Methods for Enhancing Sleep Efficiency and Prediction. *ICSMAI 2024* (Springer). DOI: 10.1007/978-3-031-66854-8_1

LANGUAGES: English (IELTS 7.0, CEFR C1) | Urdu (Native) | German (A1.2)